

The empirical recurrence for OEIS A229642, recovered from its tail

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Abstract

OEIS A229642 records “Empirical recurrence of order 69” and keeps the recurrence itself in a linked file rather than in the entry, so the conjecture cannot be read off the entry at all. It can still be settled. The entry counts arrays under a condition local to a bounded window of lines, which makes the count a walk count on a finite digraph and forces a monic integer recurrence of order at most the number $S' = 142$ of states with distinct futures. Berlekamp–Massey on enough exact terms therefore returns the minimal such recurrence, and on the tail beginning at $n = 4$, so the recurrence is proved for every $n \geq 73$. Its last coefficient is $c_{69} = -74088000 \neq 0$, so no further tail satisfies anything shorter; any order-69 recurrence the sequence satisfies from any point on must then have the same characteristic polynomial. The entry’s recurrence is the one displayed here, whatever its linked file contains, and it is proved.

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1 A conjecture that is not written down

OEIS A229642 is “Number of defective 3-colorings of a 6 X n 0..2 array connected horizontally, diagonally and antidiagonally with exactly two mistakes, and colors introduced in row-major 0..2 order.”. It has offset 1 and begins

0, 3924, 50084, 747299, 11485716, 170405645, 2495874984, 35693194243, ...

The entry states:

Empirical recurrence of order 69 (see link above).

As of the “Last modified” line on the live entry (Apr 28 2021 01:22:53, revision 10) this is still recorded as empirical, and it is still open. The coefficients are nowhere in the entry text: what is conjectured is only that a recurrence of that order exists.

2 The count is a walk count

The entry’s condition is local. It constrains a bounded window of consecutive lines of the array and nothing beyond it, so the admissible configurations of one window are the vertices of a finite digraph, an edge joins u to v when v may follow u , and an admissible array is exactly a walk. Writing M for the adjacency matrix and ι, τ for the indicator vectors of the admissible first and last windows,

$$a(n) = \iota^\top M^{j(n)} \tau$$

for a linear reindexing j fixed by the entry’s shape. States with identical futures contribute identically to that product and may be merged without changing any count; after merging $S' = 142$ states remain, and it is that bound every computation below uses.

Lemma 1. *a satisfies a monic linear recurrence with integer coefficients of order at most $S' = 142$.*

Proof. By the Cayley–Hamilton theorem $M^{S'}$ is an integer combination of $I, M, \dots, M^{S'-1}$, and multiplying that relation on the left by $\iota^\top M^j$ and on the right by τ turns it into a relation among $S' + 1$ consecutive values of a . $\square$

3 Why the question has to be asked of a tail

Running Berlekamp–Massey on the sequence from its first term returns an order LARGER than 69: the opening terms do not satisfy the recurrence, and a recurrence that holds only eventually always looks longer when the exceptional terms are included. That is not evidence against the entry. An OEIS empirical recurrence is asserted for n beyond some index, not from the first term, so the question has to be asked of the tails. It was asked of each tail in turn, and the tail beginning at $n = 4$ is the first whose minimal order is exactly the stated 69.

Lemma 2. *A sequence known to satisfy some linear recurrence of order at most S' is determined, together with its minimal recurrence, by any $2S'$ consecutive terms: Berlekamp–Massey applied to them returns that minimal recurrence exactly.*

Applied to $2S' = 284$ exact terms from $n = 4$ on, it returns order 69 — the order the entry states — with integer coefficients:

$$a(n) = \sum_{i=1}^{69} c_i a(n-i),$$

c_1	18	c_{19}	−784725533809557	c_{37}	7428981351257592438	c_{55}	12505432818446247
c_2	219	c_{20}	−683028562217688	c_{38}	−12090750754169910345	c_{56}	8972100216174972
c_3	−4476	c_{21}	5461265454862252	c_{39}	−6173105033527396919	c_{57}	−2439190763366325
c_4	−24111	c_{22}	2736837151583931	c_{40}	12976419886356629034	c_{58}	−1266026154204738
c_5	479154	c_{23}	−30193305859911384	c_{41}	3762064540632527481	c_{59}	357683471229951
c_6	1779214	c_{24}	−5089179639449873	c_{42}	−11184550258379536189	c_{60}	134624240826535
c_7	−28772160	c_{25}	132711444265371534	c_{43}	−1458950193753196533	c_{61}	−38807265792690
c_8	−88762209	c_{26}	−16541761371220662	c_{44}	7774201898984887125	c_{62}	−10389031526286
c_9	1099386674	c_{27}	−463709261583841589	c_{45}	96467469861142608	c_{63}	3018757395240
c_{10}	2988194793	c_{28}	177503203029364737	c_{46}	−4365111119017228761	c_{64}	547133196420
c_{11}	−28920237897	c_{29}	1287659528044572096	c_{47}	332249359106069952	c_{65}	−159052051656
c_{12}	−69230163869	c_{30}	−792020970894412371	c_{48}	1978864436380569681	c_{66}	−17532713304
c_{13}	553030442322	c_{31}	−2840143719539270037	c_{49}	−282406029449333421	c_{67}	5078006640
c_{14}	1131387620874	c_{32}	2369586944417149242	c_{50}	−722373667717373358	c_{68}	257191200
c_{15}	−7986018135403	c_{33}	4968222313309183867	c_{51}	139010670948666025	c_{69}	−74088000
c_{16}	−13291193976018	c_{34}	−5252790921833927907	c_{52}	211275305689627056		
c_{17}	89296296834612	c_{35}	−6865656905302830228	c_{53}	−48348586698902868		
c_{18}	112940360024314	c_{36}	8980510303650347943	c_{54}	−49118707489275568		

4 The recurrence, and the identification

Theorem 3. *The recurrence above holds for every $n \geq 73$, and it is the recurrence OEIS A229642 records.*

Proof. That it holds is Lemma 2 applied to the tail from $n = 4$, whose minimal recurrence it is.

For the identification, write $q_{\min}$ for the characteristic polynomial of that minimal recurrence, $x^{69} - c_1x^{68} - \dots - c_{69}$. Its constant term is $-c_{69} = 74088000$, which is not zero, so $q_{\min}(0) \neq 0$ and $q_{\min}$ has no root at the origin. A solution of a constant-coefficient recurrence is a combination of terms $n^k \lambda^n$ with λ a root of its characteristic polynomial, and only the components with

$\lambda = 0$ vanish after finitely many terms. Since $q_{\min}$ has none, no component of a dies out, and the minimal recurrence of every further tail is again $q_{\min}$ — the order cannot drop below 69 at any later starting point.

Now let q be the characteristic polynomial of whatever recurrence the entry asserts. The entry asserts it has order 69, so $\deg q = 69$, and it is monic. It holds from some index t on. From $\max(t, 4)$ on, the sequence satisfies both q and $q_{\min}$, and the minimal polynomial of that tail is $q_{\min}$ by the previous paragraph; therefore $q_{\min} \mid q$. Both are monic of degree 69, so $q = q_{\min}$. $\square$

5 Verification

Four checks, all in exact integer or rational arithmetic.

First, the walk model reproduces all 15 terms the entry publishes, exactly. This is what ties the model to the entry, and it is the only place where reading the entry’s English enters the argument.

Second, the order was computed twice by independent routes: once modulo a 61-bit prime, where every intermediate is a machine word, and once over $\mathbb{Q}$ in exact rational arithmetic. Both return 69, and the exact run additionally confirms every coefficient is an integer — had any been a proper fraction the recurrence would not be the entry’s.

Third, the recovered recurrence was evaluated directly on the entry’s own published terms, with no matrices and no Berlekamp–Massey involved. It holds at each of the 0 indices where the entry publishes enough earlier terms to test it.

Fourth, the last coefficient was checked to be nonzero, which is what the identification above turns on; it is -74088000 .

Remark 4. The conjecture’s own statement was never read. The entry pins it down to the family of order-69 recurrences this sequence satisfies from some point on, and that family turns out to have exactly one member.

References

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